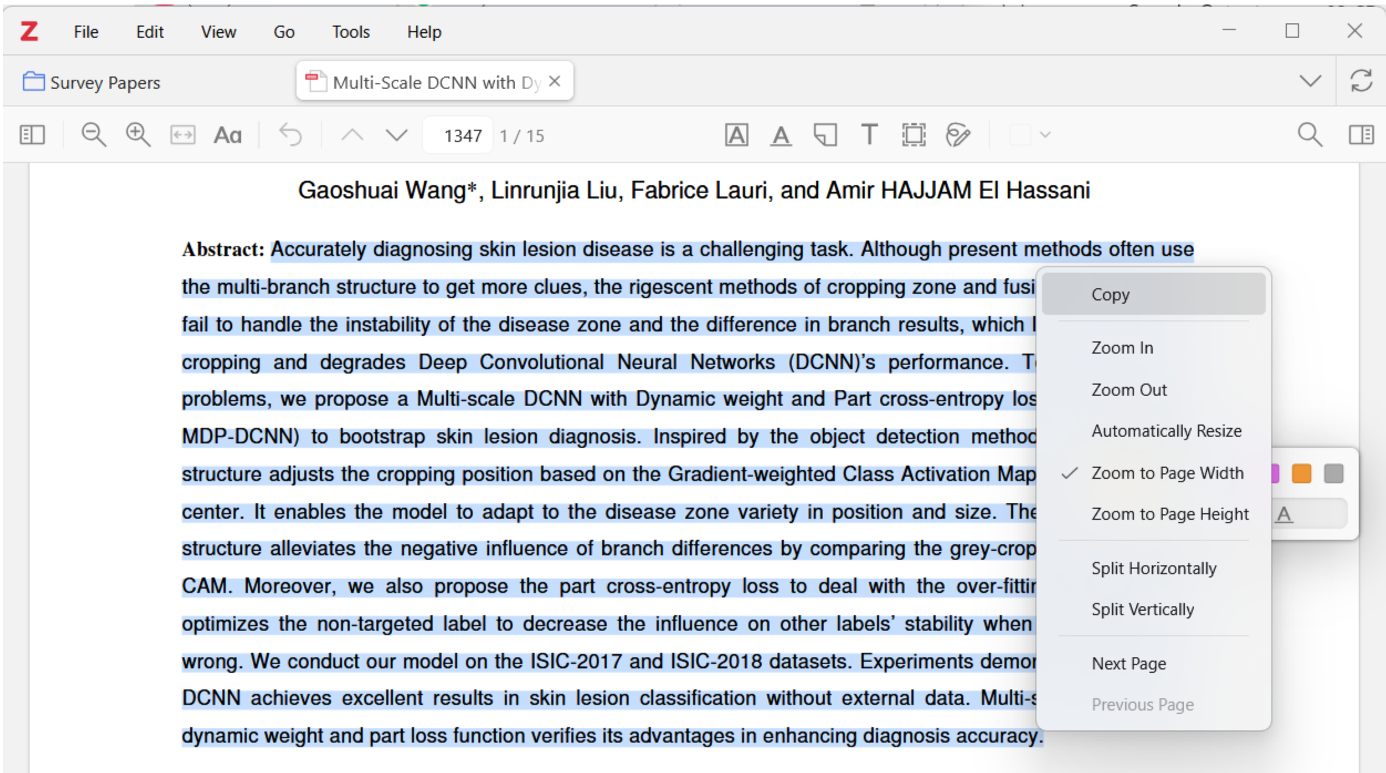
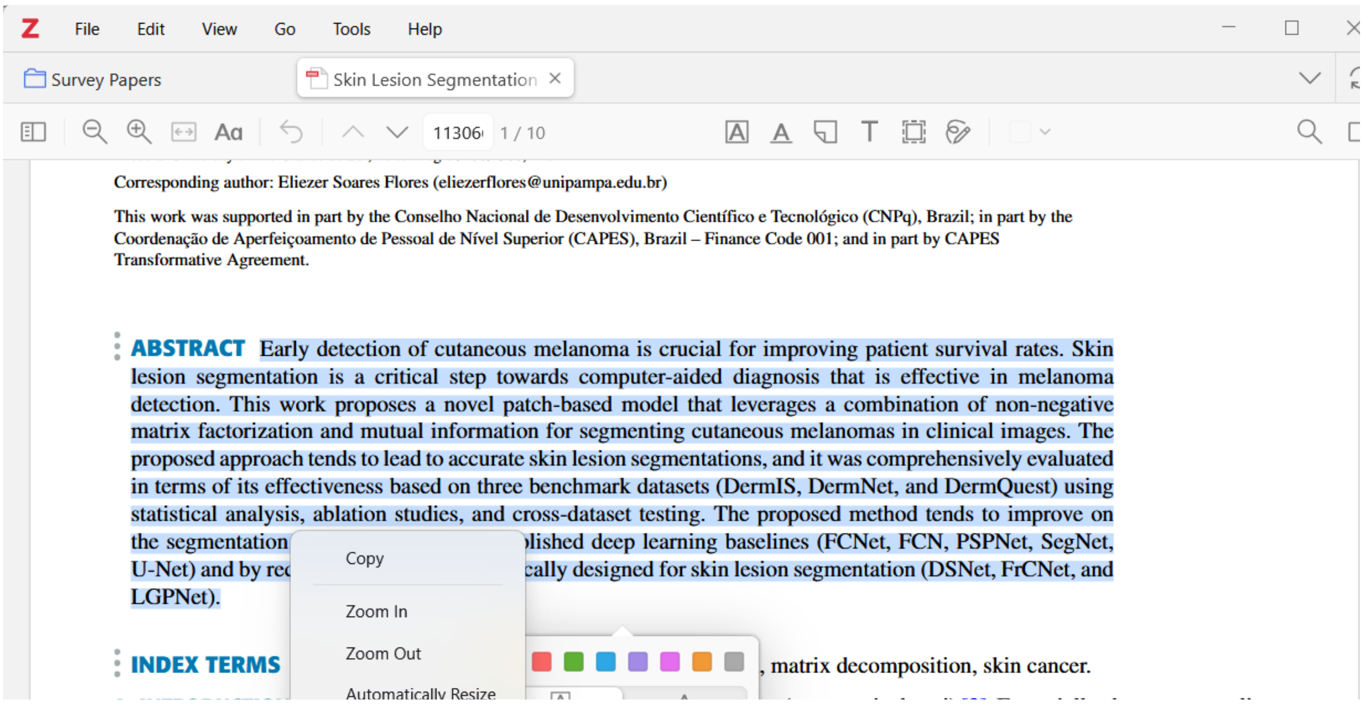
Assignment 2: Literature Mining using ChatGPT API

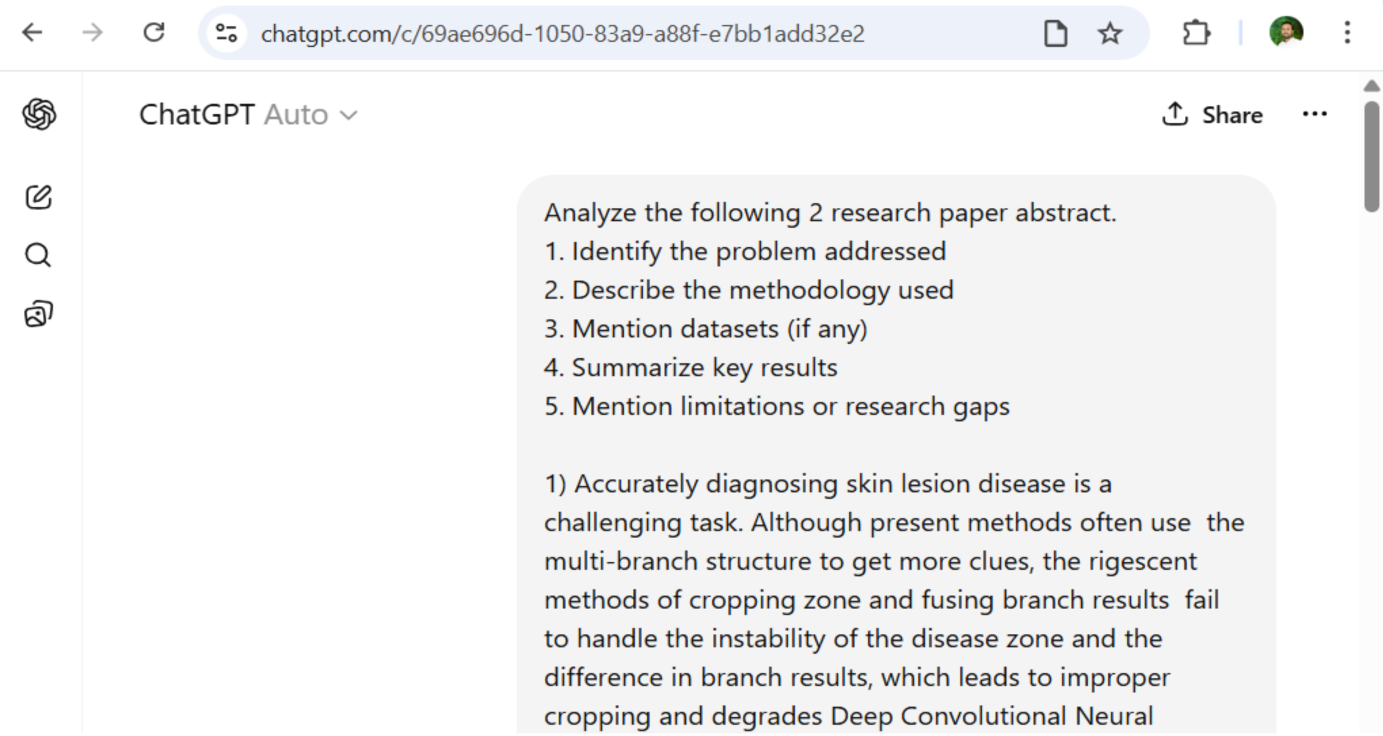
Step 1: Extracting the First Abstract



Step 2: Extracting the Second Abstract



Step 3: Structuring the AI Prompt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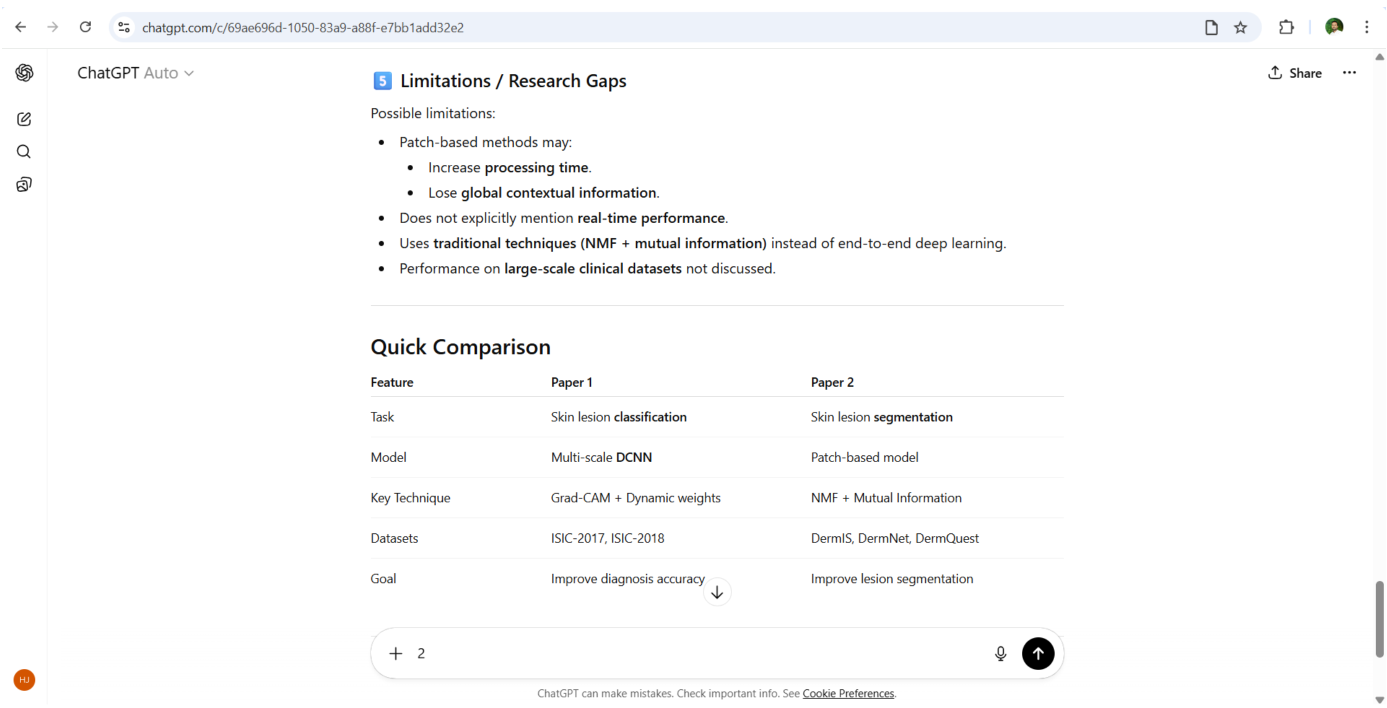
Analyze the following 2 research paper abstract.
1. Identify the problem addressed
2. Describe the methodology used
3. Mention datasets (if any)
4. Summarize key results
5. Mention limitations or research gaps

1) Accurately diagnosing skin lesion disease is a challenging task. Although present methods often use the multi-branch structure to get more clues, the rigescent methods of cropping zone and fusing branch results fail to handle the instability of the disease zone and the difference in branch results, which leads to improper cropping and degrades Deep Convolutional Neural

Step 4: Generating the Comparison and Analysis


chatgpt.com/c/69ae696d-1050-83a9-a88f-e7bb1add32e2

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5 Limitations / Research Gaps

Possible limitations:

- Patch-based methods may:
  - Increase **processing time**.
  - Lose **global contextual information**.
- Does not explicitly mention **real-time performance**.
- Uses **traditional techniques (NMF + mutual information)** instead of end-to-end deep learning.
- Performance on **large-scale clinical datasets** not discussed.

Quick Comparison

Feature	Paper 1	Paper 2
Task	Skin lesion **classification**	Skin lesion **segmentation**
Model	Multi-scale **DCNN**	Patch-based model
Key Technique	Grad-CAM + Dynamic weights	NMF + Mutual Information
Datasets	ISIC-2017, ISIC-2018	DermIS, DermNet, DermQuest
Goal	Improve diagnosis accuracy	Improve lesion segmentation

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